

Mohammadreza Teymoorianfard

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Research Summary

Ph.D. candidate at UMass Amherst working on AI security and safety, with a focus on embodied and agentic systems. My research studies vulnerabilities in reasoning-enabled AI systems, autonomous agents, and generative models. My work spans adversarial evaluation, privacy, and provenance, with publications at USENIX Security and collaborations with Qualcomm and Cisco.

Interests: AI Security & Safety · Embodied AI · AI Agents · Multimodal/VLA Models

Education

University of Massachusetts Amherst

PH.D. IN COMPUTER SCIENCE

Sep 2023 – Expected 2028

- Advisor: Prof. Amir Houmansadr | GPA: 3.88/4

University of Massachusetts Amherst

M.SC. IN COMPUTER SCIENCE

Sep 2023 – May 2025

- Advisor: Prof. Amir Houmansadr | GPA: 3.87/4

University of Tehran

B.SC. IN ELECTRICAL ENGINEERING, MINOR IN COMPUTER ENGINEERING

Sep 2018 – Jun 2023

- GPA: 19.06/20 (4/4) | Ranked 3rd of 120 students

Publications

Network-Level Prompt and Trait Leakage in Local Research Agents

H. Jeong, **M. Teymoorianfard**, A. Kumar, A. Houmansadr, E. Bagdasarian

USENIX Security '26. [Paper] [Code]

Visual Grounding and Deliberation Help Multi-Robot Teams Make High-Stakes Decisions

M. Teymoorianfard, S. Mahmud, J.-P. Monteuuis, J. Petit, S. Zilberstein, A. Houmansadr, E. Bagdasarian

Under review, 2026.

Rethinking Image Watermarking in the Era of Frontier Generative Models

A. Naseh, **M. Teymoorianfard**, A. Houmansadr

Under review, 2026.

ReasonBreak: Probing Vulnerabilities in Reasoning-Enabled Vision-Language-Action Models for Autonomous Driving

M. Teymoorianfard, J.-P. Monteuuis, J. Petit, A. Houmansadr

arXiv preprint, 2026. [Paper]

TRACE: Trajectory Reasoning through Adaptive Cross-Step Evidence Aggregation for LLM Agents

V. Mittapalli, S. J. Dani, S. S. Pilli, S. Ansu, **M. Teymoorianfard**, F. Dernoncourt, H. Chen, Y. Wang, R. A. Rossi, N. K. Ahmed

arXiv preprint, 2026. [Paper]

VidStamp: A Temporally-Aware Watermark for Ownership and Integrity in Video Diffusion Models

M. Teymoorianfard, S. Sitaraman, S. Ma, A. Houmansadr

arXiv preprint, 2025. [Paper] [Code]

MeanSparse: Post-Training Robustness Enhancement Through Mean-Centered Feature Sparsification

S. Amini, **M. Teymoorianfard**, S. Ma, A. Houmansadr

arXiv preprint, 2024. [Paper] [Code]

Experience

UMass Amherst — The Secure, Private Internet (SPIN) Research Group

GRADUATE RESEARCH ASSISTANT

Sep. 2023 – Present

- **Multi-Robot Safety Decisions (Qualcomm collaboration)**
Benchmarked VLM robot teams on safety decisions under distributed information and designed HUDDLE, a deliberation protocol improving team accuracy by up to 42 percentage points (to 96%).
- **ReasonBreak — Security of Reasoning-Enabled Driving VLAs (Qualcomm collaboration)**
Designed and implemented a black-box evaluation framework for vulnerabilities in reasoning-enabled driving VLAs (NVIDIA Alpamayo, AutoVLA); attacks reached up to 100% reasoning ASR and 72% closed-loop trajectory ASR.
- **Privacy of Local Research Agents (USENIX '26)**
Showed local research agents leak user prompts and traits from network metadata alone, and proposed a synthetic-query defense.
- **Watermarking for Video Diffusion Models (VidStamp)**
Designed a decoder-embedded, frame-level watermark enabling high-capacity, imperceptible marking and temporal tamper localization.
- **Image Watermarking vs. Frontier Models**
Developed an API-only regeneration attack achieving 89% mean watermark removal across 13 watermarking methods, and proposed K-SemMark-R, a registry-based content watermark resistant to it.
- **Adversarial Robustness (MeanSparse)**
Co-developed a post-training feature-sparsification method with state-of-the-art AutoAttack accuracy on CIFAR-10/100 and ImageNet at release.

UMass Amherst Industry Data Science Practicum (with Cisco)

RESEARCH MENTOR

Spring 2026

- Mentored a four-person M.S. student team on an industry research project with Cisco, from problem formulation through publication. The resulting TRACE agent-monitoring method reached 0.844 recall on SHADE-Arena while using 35% fewer LLM calls than prior monitors.

University of Tehran

UNDERGRADUATE RESEARCHER

Jun. 2021 – Jun. 2023

- Model Predictive Control for autonomous taxi navigation (Smart Networks Lab, advisor: Hamed Kebriaei); text detection system for card images (advisor: Reshad Hosseini).

Honors & Awards

- 2026 **Edward Riseman and Allen Hanson Scholarship** *UMass Amherst CICS*
- 2026 **GREPSEC VII and USENIX Security '26 Student Grants** *USENIX Association*
- 2023 **Industry-sponsored award for outstanding bachelor's thesis** *University of Tehran*
- 2019 – 2020 **Sponsors Foundation and Faculty of Engineering top-student awards** *University of Tehran*
- 2018 **Top 0.4% of 150,000+ students, National University Entrance Exam**

Academic Service & Activities

- **Mentoring:** supervised 12 undergraduates (4 cohorts, UMass URV program), each presenting a research poster
- **Reviewing:** sub-reviewer for ICLR 2026, CVPR 2026, ECCV 2026, and NeurIPS 2026
- **Teaching:** Teaching Assistant at UMass Amherst for 7 semesters — Machine Learning and Intro to Programming courses
- **Presentation:** Poster, New England Security Day (NESD), 2026

Technical Skills

ML Systems: PyTorch, Hugging Face Transformers, scikit-learn, NumPy

Programming: Python, C/C++, $\text{\LaTeX}$, MATLAB

Simulation & Infrastructure: AlpaSim, Isaac Sim, Slurm, Apptainer/Docker, Linux, Git